

CryptoAgent Project Part A: Data Design and Analysis

Report*

Kengtao Wu

July 24, 2025

Abstract

The cryptocurrency market has evolved into a significant asset class characterized by extreme volatility, rapid innovation, and a 24/7 global trading cycle. The interest in cryptocurrency from investors has been rising. Therefore, this paper outlines the design and implementation of the data and feature engineering pipeline for the product CryptoAgent. In stage 1, CryptoAgent transitions from a single-provider ETL process to a multi-exchange aggregation framework, aiming to overcome the problems of 'aggregate confusion' identified in G. Schwenker's study. In feature engineering (stage 2), I refine the existing factors and introduce new factors derived from socially validated narrative pipelines by weighting the output of a rule-based sentiment model with a score computed from the engagement of the social media platform (X / Twitter).

Keywords: Cryptocurrency, Quantitative Factor Investing, Sentiment Analysis, Data Reconciliation, Social Media Validation

1 Structured Data

1.1 Aggregate confusion in Crypto Market

The performance of any quantitative strategy depends on the quality of its input data. First and foremost is the single-source fallacy. The standard approach to acquiring crypto data is to use a data aggregator API such as CoinDesk or CryptoCompare. However, this might lead to implicit trust in a third party’s black-box data processing. Research conducted by Schwenkler et al. 2025 on “aggregate confusion, states that there are two systemic errors happening in the current market. (1) study found that as 21 percent of all coins may be subject to undisclosed ID changes, and CoinGecko reuses the same ID for distinct cryptocurrencies in 16 percent of their sample. (2) Daily closing prices and trading volumes vary enormously between providers. Deviations in daily close prices can be as large as a factor of (e.g., on Coinpaprika), while volume data is even noisier. (3) Industry systemic risk covers all major aggregators, such as CoinMarketCap, CoinGecko, CryptoCompare, etc. Therefore, the primary and most important task is to build a reliable and audit data acquisition and cleansing process.

1.2 Multi-Source Verification and Reconciliation Pipeline

To counter the fragility of a single data source, CryptoAgent shifts from a single API call to a multi-source data ingestion and reconciliation process. In CryptoAgent, multisource data can be achieved using standardized Python libraries, such as CryptoCurrency eXchange Trading (cctx). cctx is a standardized tool that provides a unified API interface to these and more than 100 other exchanges. This allows us to bypass the aggregators’ black-box processing and access the first-hand streams from the exchanges themselves directly. We have complete control over the entire data reconciliation and sanitization workflow. CryptoAgent fetches data directly from the 5 selected multiple exchanges. (Binance, Coinbase, Kraken, Bitfinex, OKX) CryptoAgent co-opts the arbitrageur’s logic and turns this market inefficiency into a strength for our structured data pipeline. Specifically, our aggregation algorithm acts as an ‘arbitrageur’, not for profit but for the purpose of constructing a unified and accurate price series by identifying and reconciling these price differences.

1.3 Two-step clustering approach

Consider that simply averaging data across different 5 exchanges might remain highly susceptible to outliers. Therefore, to construct a reliable consensus, I implemented an advanced aggregation algorithm based on the idea in Schwenkler et al.’s study. The idea is based on a principle akin to zero-knowledge proof, assuming that we do not know a priori which exchange is correct, but utilize the algorithm to identify the consensus and programmatically discard dissenters.

The first challenge is to ensure that all three of our sources report data for the same underlying cryptocurrencies, for a given ticker (e.g., AMB, which could refer to AirDAO or Ambrosus). Therefore, CryptoAgent pulls the historical market capitalization data from each exchange, then applies

a clustering algorithm to group the data stream. If all the sources are in one cluster and within 8 percent tolerance threshold ($\text{tol} = 8\text{percent}$), it means they have reached a consensus identification. In contrast, any exchange whose reported price deviates from this median by more than the predefined tolerance thresholds will be flagged as an outlier and excluded from the consensus group for specific cryptos. As a result, data from a consistent set of exchanges is passed to step 2.

In step 2, closing price clustering, we repeat the clustering process on the historical close prices of those data that have passed step 1, applying a stricter tolerance threshold of $\text{tol} = 2$ percent to filter out exchanges that identify the correct cryptocurrency but report the incorrect price. Since exchanges may experience faults, API problems, or delays in price movement. As a result, only the data that fall within the tight consensus pool are included in the final step.

In final step median aggregation, within the largest consensus provider cluster, we use the median to aggregate all the market metrics (Open, High, Low, Close, Volume). Allouche et al. (2023) state that cryptocurrency data exhibit a tailed distribution. Therefore, the median is a more appropriate way than the mean since it is not skewed by single extreme outliers.

1.4 Volume adjustment

Consider that over 70 percent of reported trading volume on some unregulated exchanges is fake, generated by "wash trading," where entities trade with themselves to create a misleading impression of liquidity. The latest wash trading detection methods, such as Benford's law analysis, often require granular, tick-level data and complex statistical tests, which are beyond the scope of our project. Therefore, I construct our final adjusted volume metrics by weighting to the exchanges. In detail, these weights are based on the reputation and historical data quality of each exchange. For the project, I assign the following weights: Binance (1.0), Coinbase (0.95), Kraken (0.9), Bitfinex (0.8), and OKX (0.85). As a result, we might downweight data from potentially less reliable sources, providing more accurate input for Station 2. It should be noted that the exchange weights are assigned based on subjective judgment rather than strong evidence. My approach cannot approach temporal variations; lawful trading may be mistakenly classified as suspicious, or vice versa.

Table 1: Descriptive Statistics Comparison: Raw vs. Aggregated Data

Metric (BTC/USD)	Data Source	Mean	Median	Std. Dev.	Skewness	Kurtosis
2^* Daily Log Return	Single Source (CryptoCompare)	0.0013	0.0006	0.034	-1.9	34.5
	Aggregated Data	0.0017	0.0003	0.025	0.2	6.6
2^* Daily Log Volume	Single Source (CryptoCompare)	23.3	23.4	0.7	-0.5	3.1
	Aggregated Data	8.9	9.6	2.5	-0.4	2.2

2 Unstructured Data

2.1 Data Source Architecture

In the cryptocurrency market, where narratives drive a significant portion of price action, the quality and reliability of news data are crucial. A simplistic method is to trust every article from CoinDesk, but it might miss a fact; not all news has the same impact or credibility. Therefore, I've developed a validation pipeline for Station 1. The function collects and cleans news data from CoinDesk and integrates a socially driven mechanism for narrative validation. The core idea is to use the social media platform (from X/Twitter) as a real-time filter to assess and score the relevance and trustworthiness of formal news articles sourced from CoinDesk. Consider January 2024, when CoinDesk reported Bitcoin ETF approvals by the SEC. Immediately, Twitter exploded with both voices, whether confirming or rejecting, which proved a real-time credibility assessment beyond sentiment analysis. This case highlights that social media might act as an editorial board, offering instant fact-checking and prioritizing information. Moreover, I was inspired by Surowiecki's Wisdom of Crowds theory. In centralized traditional finance, information quality is vetted by credit rating agencies or the government. In contrast, in the decentralized cryptocurrency market, this role is fulfilled by a distributed network of thousands of engaged and financially incentivized participants, including traders, developers, journalists, and enthusiasts. My idea is to treat the aggregate social media response not as sentiment, but as a judgment of credibility and importance of a given news.

2.2 Data Processing and Validation

Our pipeline is built on two parallel data streams, designed to capture both the formal, structured narrative of the market and its raw, real-time data. News articles are retrieved from CoinDesk, and the Twitter API is utilized to gather social signals. Second, CryptoAgent uses BeautifulSoup to parse the raw HTML and extract the body of the article, discarding advertisements and other elements. Furthermore, remove URLs, email addresses, and special characters, and convert to lowercase and Unicode normalization. Third, for the news event we collected, I applied a corresponding query to the Twitter data stream, by calculating three metrics: (1) Engagement Velocity, which represents a way to measure how quickly the market is paying attention to the news.

$$EV = \frac{\text{Total engagement (Likes + Retweets) in first 60 mins}}{60}$$

For example, a high EV (above 100 engagements/minute) might represent a high-impact news story that has captured the market's attention and is spreading rapidly. Instead, a low EV lower than 20 suggests the news is either niche or simply noise. Which effectively acts as a filter that prioritises what truly matters. Furthermore, consider that not all opinions on Twitter are equal in importance. Thus, the CWS is designed to measure the sentiment of discourse among certified, credible accounts and filter out noise from unverified users, bots, and other sources. I use the square root of the follower

count to minimize the effect of mega-influencers (e.g., Elon Musk’s account or Trump’s account). This enables our CWS to be more influenced by a consensus among all group members, regardless of their size. A positive CWS might indicate that credible voices are reacting in the market, while a negative CWS might represent expert concern or scepticism.

$$\text{Weight}_i = \sqrt{\text{Followers}_i} (1 + I_{\text{verified}, i})$$

In addition, consider that there can often be a disconnect between the news and social media sentiment. To address this, the Sentiment Alignment Score (SAS) measures the difference between the VADER compound score of the original CoinDesk news and our CWS derived from Twitter. This final metric captures the alignment between narrative and market perception.

$$\text{SAS} = 1 - |\text{VADER}_{\text{Article}} - \text{CWS}|$$

2.3 Final Quality Score and Data Output

While each metric’s Engagement Velocity (EV), Credibility-Weighted Sentiment (CWS), and Sentiment Alignment Score (SAS) provides insight into the narrative’s market reception, we still require a single, comprehensive Quality Score. In detail, I assign the highest weight, 40 percent to EV because informational importance is a prerequisite for impact. Second, weight 30 percent to the Sentiment Alignment Score (SAC) and weight 30 percent to the Consensus Component, which ranges from [-1, 1], which means only incorporate the positive sentiment from credible sources. Although negative sentiment might be valuable in risk analysis, it is not a component of “quality” in this specific score.

$$\text{Quality score} = 0.4 \text{ norm}(\text{EV}) + 0.3[\text{SAS} + \text{ReLU}(\text{CWS})].$$

Table 2: Cryptocurrency News Quality Scores

Headline	EV (eng/min)	CWS	SAS	Final Quality Score
High Score Examples				
“SEC Approves Spot Bitcoin ETFs in Landmark Decision”	152.8	0.85	0.90	0.91
“Bitcoin Bulls ‘Under-positioned’ as Trump Pressure on Fed Drives 15B Into BTC ETFs”	121.5	-0.75	0.88	0.88
Low Score Examples				
“New Meme Coin ‘GalacticShiba’ Announces Airdrop for Early Adopters”	3.2	0.45	0.70	0.23
“GSR Leads 100M Private Placement Into MEI Pharma to Launch First Institutional Litecoin Treasury Strategy”	1.5	0.95	0.98	0.15

To illustrate how our pipeline distinguishes between high-impact signals and low-value noise, we present several news headlines along with their corresponding computed metrics. It should be noted that the tab.2 is based on simulated data, which allows us to clearly demonstrate the logic of our scoring mechanism. It ensures that our final quality-weighted sentiment factor, which we built in Station 2, is driven by news verified by the community, rather than low-value or non-relevant content. Therefore, our station 1 can separate the market-moving signals from the noise, rather than collecting and cleaning the data.

3 Feature Engineering

The objective of Station 2 is to transform the clean data from Station 1 into a predictive factor derived from both structured market/on-chain data and unstructured news sentiment data. My approach emphasizes simplicity and practical utility. I aim to create a streamlined set of features that capture the main drivers of cryptocurrency price movements.

3.1 The Momentum Factor

According to Jagadeesh and Titman (1993), assets that have performed well recently tend to keep performing well in the near term; vice versa, underperformers tend to remain weak. This trend-following behaviour is especially pronounced in the crypto market, where investor sentiment and narratives can drive price movement. Furthermore, in accordance with academic methodology, CryptoAgent calculates the momentum for each cryptocurrency asset (i) at the beginning of the week (t) using a 63-day look-back period. First, we calculate the total return over the period from 13 weeks ago to 1 week ago. Importantly, we ignore the most recent week’s return, as well as the 5 trading days, because assets that have experienced a massive spike in the last few days are often due for a short-term pullback.

$$\text{MOM}_{63d_{i,t}} = \text{Total Return from week } t - 13 \text{ to week } t - 2$$

Every week, we rank all cryptos by this score, then form a portfolio. We buy the top 20 percent of assets for the long portfolio and sell the bottom 20 percent of assets for the short portfolio. The return of this long-short portfolio is our momentum factor, therefore, this factor should have a positive average return over time.

3.2 The Short-Term Reversal Factor

In contrast to momentum, Short-Term Reversal operates on the opposite principle over a shorter horizon. (e.g., 1 to 4 weeks) We assume that when cryptos experiencing price movements in the immediate past are likely to “revert” or move in the opposite direction in the subsequent periods. As theorized by De Bondt (1985), investors tend to overreact to news; sudden positive news can lead to euphoric buying, driving asset prices far above their fundamental value. Conversely, negative news

might lead to panic selling. Therefore, STR strategy can be explained by a bet that these emotionally driven overreactions are temporary, and the price will eventually mean revert. The cryptocurrency market is dominated by retail investors who are more susceptible to behavioral biases, such as FOMO (Fear of Missing Out) and FUD (Fear, Uncertainty, and Doubt). At the start of each rebalancing week t , for every cryptocurrency i , we calculate its total return over the previous trading week (from the close of week $t-2$ to the close of week $t-1$). Which can be explained by the following equation:

$$\text{STR_7d}_{i,t} = \left(\frac{P_{i,t-1}}{P_{i,t-2}} \right) - 1$$

CryptoAgent ranks all assets based on their STR 7 days score, from lowest to highest. Then divide the ranked cryptos into five equal-sized portfolios and buy an equal weight of these assets in the bottom, which were last week’s biggest losers. We are betting that these cryptos will bounce back. On the other hand, we are selling an equal portfolio of the top-ranked assets, which were the biggest winners last week. We are betting that these cryptos will pull back.

3.3 Engineering Narrative Factors

The original news sentiment pipeline involves a three-step process. First, raw text is processed through an NLP pipeline, which includes tokenization, stop-word removal, and lemmatization. Second, VADER is applied to the cleaned text to generate a score (negative, neutral, positive, or compound) for each news, with binary classification based on a compound score greater than 0.05. However, not every news is not equally important and credible; its inability to distinguish between high-quality signal and noise might reduce the sentiment factor’s predictive power. Thus, I address this limitation by incorporating the Quality Score from station 1 as article weights rather than treating all news sources equally. For cryptos i on day t , the Quality-Weighted Sentiment (QWS) is:

$$\text{Daily QWS}_t = \frac{\sum_{j=1}^{N_t} (\text{VADERScore}_j \times \text{QualityScore}_j)}{\sum_{j=1}^{N_t} \text{QualityScore}_j}$$

Where VADER Score(j) is the compound sentiment score for news j and Quality Score(j) is the composite validation score from Station 1. This ensures that high-quality news has greater influence on the final sentiment measures. We enhance traditional sentiment analysis by incorporating article quality weighting. Instead of treating all news sources equally, we weight each article’s sentiment score by a composite quality metric based on source credibility and market engagement. This addresses the signal-to-noise problem in financial news analysis.

Furthermore, although the Quality-weighted sentiment factor can capture the direction of the market narrative, it’s equally important to measure its intensity. Ngelberg and Gao (2011) demonstrate that a rise in attention might lead to price pressure and predictable returns patterns. Thus, we

construct a News Buzz metric defined as the sum of the Quality Scores of all validated CoinDesk headlines for asset i on day t :

$$\text{News_Buzz}_{i,t} = \sum_{j=1}^{N_{i,t}} \text{Quality_Score}_j$$

In simple terms, New Buzz simply sums up the quality score of all its related articles for that day. As a bigger sum means getting more high-credibility airtime. While high buzz does not always mean the price will go up, buzz factor provides a view to understanding other signals. For instance, strong momentum signals may become more convincing when they're also backed by fuzz factor. In contrast, a high buzz score without a clear price could be a warning sign of uncertainty.

4 References

- Bali, T.G., Cakici, N. & Whitelaw, R.F. (2011). Maxing out: Stocks as lotteries and the cross-section of expected returns. *Journal of Financial Economics*, 99(2). <https://doi.org/10.1016/j.jfineco.2010.08.014>
- Cong, L.W., Li, Y. & Wang, N. (2021). Tokenomics: Dynamic Adoption and Valuation. *The Review of Financial Studies*, 34(3). <https://doi.org/10.1093/rfs/hhaa089>
- Da, Z., Engelberg, J. & Gao, P. (2011). In Search of Attention. *The Journal of Finance*, 66(5). <https://doi.org/10.1111/j.1540-6261.2011.01679.x>
- De Bondt, W.F.M. & Thaler, R. (1985). Does the Stock Market Overreact? *The Journal of Finance*, 40(3). <https://doi.org/10.1111/j.1540-6261.1985.tb05004.x>
- Jegadeesh, N. & Titman, S. (1993). Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency. *The Journal of Finance*, 48(1). <https://doi.org/10.1111/j.1540-6261.1993.tb04702.x>
- Maître, A.T., Pugachyov, N. & Weigert, F. (2024). Twitter-Based Attention and the Cross-Section of Cryptocurrency Returns. *SSRN Electronic Journal*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5010136
- Schwenkler, G., Shah, A. & Yang, D. (2025). Aggregate Confusion In Crypto Market Data. *SSRN Electronic Journal*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4673905
- Surowiecki, J 2004, *The Wisdom of Crowds: Why the Many Are Smarter Than the Few and How Collective Wisdom Shapes Business, Economies, Societies, and Nations*, Doubleday, New York

5 Appendix

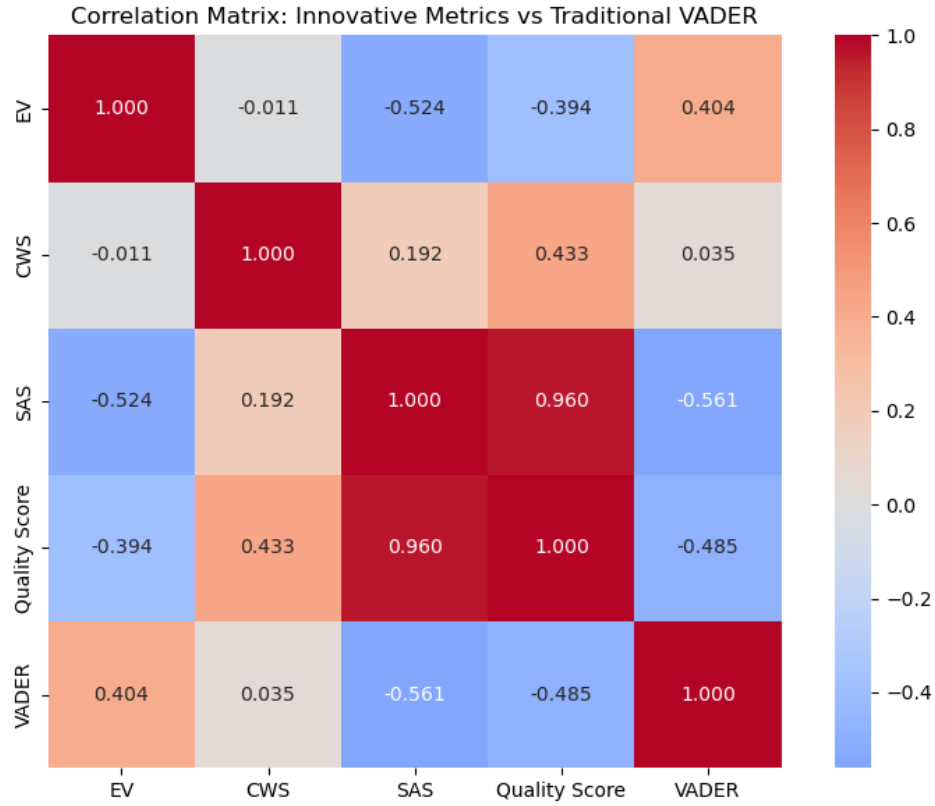


Figure 1: shows the correlation between the proposed twitter validation metrics and traditional VADER sentiment scores. The sentiment alignment score and quality score present very high correlation 0.96, indicating that our metric captures sentiment alignment. Both Engagement Velocity and Quality Score show negative correlations with VADER compound scores (-0.394 and -0.485 respectively), suggesting that traditional sentiment analysis may miss important market dynamics captured by social media engagement.

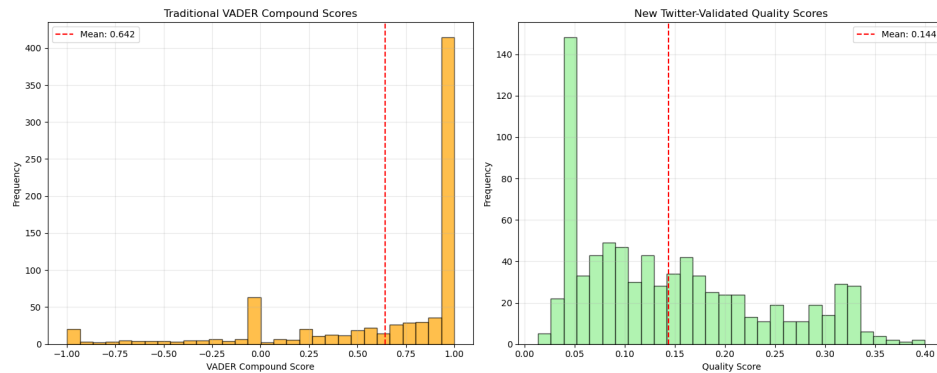


Figure 2: shows the compares the distribution of traditional VADER compound scores and our validated quality score. VADER shows right skewed with mean 0.642, while quality score shows mean number of 0.144, indicates more strict assessment of news quality, with better separation between high and low quality news.

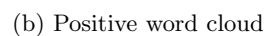
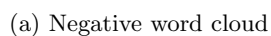


Figure 3: Negative and positive word cloud